Project 4 Description

<u>Investigation of Variable Selection Algorithms</u>: The goal of this project is to investigate different model selection techniques in linear regression. We will investigate choosing the best fitting model using backward selection based on F-tests and p-values (automated), AIC, BIC, lasso, and elastic net. The cases to investigate are 1) N=250 subjects with 20 variables in the model. 5 of them would be considered important with coefficients between 0.5/3 and 2.5/3 in 0.5/3 unit increments (0.5/3, 1/3, 1.5/3, 2.0/3, and 2.5/3). In case 1a consider all X variables as independent from one another. In case 1b, induce correlation between the X variable (we'll have to specify how as a group). 2) N=500 subjects with 20 variables in the model. 5 of them would be considered important with coefficients between 0.5/3 and 2.5/3 in 0.5/3 unit increments (0.5/3, 1/3, 1.5/3, 2.0/3, and 2.5/3). In case 2a consider all X variables as independent from one another. In case 2b, induce correlation between the X variable (we'll have to specify how as a group).

For each case first simulate the data using the hdrm package that can be found on github. Then fit each model using the 5 model fitting approaches above. There are different choices for how to choose lambda for lasso and elastic net. Investigate at least 2 choices (we'll have to specify the choices as a group). Summarize your findings in terms of true positives (what percentage of the time are variables X1 - X5 in the final model) and false positive rates (confusion matrix is another name for this). In other words, how often does your final model include any of the 15 variables that are not associated with the outcome and how often does your final model include each of the 5 variables with a known relationship with Y. For the variables that are retained in the model, what is the bias of the parameter estimates, what is the coverage of the 95% CI, and what is the type I and II error of the variables selected. For the variables retained, you can do testing on the final model (in other words refit the final model like no model selection was done) or propose an alternative approach to testing since we believe there is bias in just fitting the final model like there was no variable selection done.

I will post papers that may be of help. A google scholar search will identify many more papers that could be of use I am sure.